MATH 1075 - Section 9.5 Worksheet

10/29/2026

1 Graphing Linear Inequalities in Two Variables

Graphing a Linear Inequality in Two Variables

Step 1 Write the inequality with the y variable isolated, if possible.

Step 2 Graph the related equation. Draw a dashed line if the inequality is strict, $<$ or $>$. Otherwise, draw a solid line.

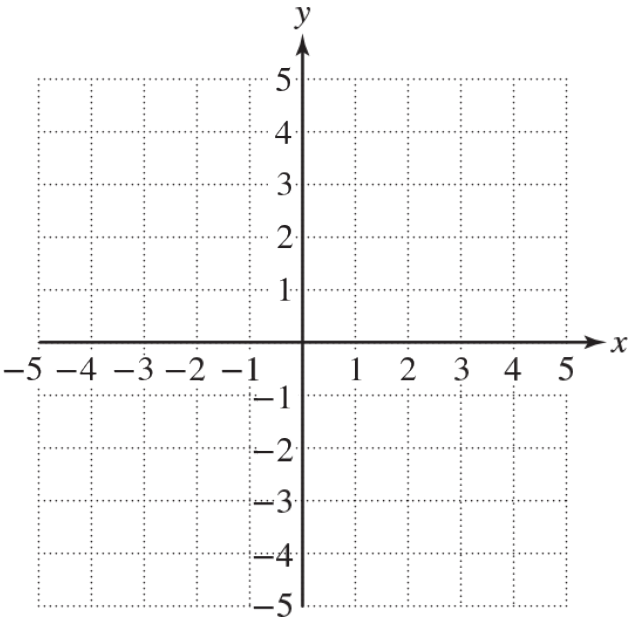
Step 3 Shade above or below the line as follows:

- Shade *above* the line if the inequality is of the form $y > mx + b$ or $y \geq mx + b$.
- Shade *below* the line if the inequality is of the form $y < mx + b$ or $y \leq mx + b$.

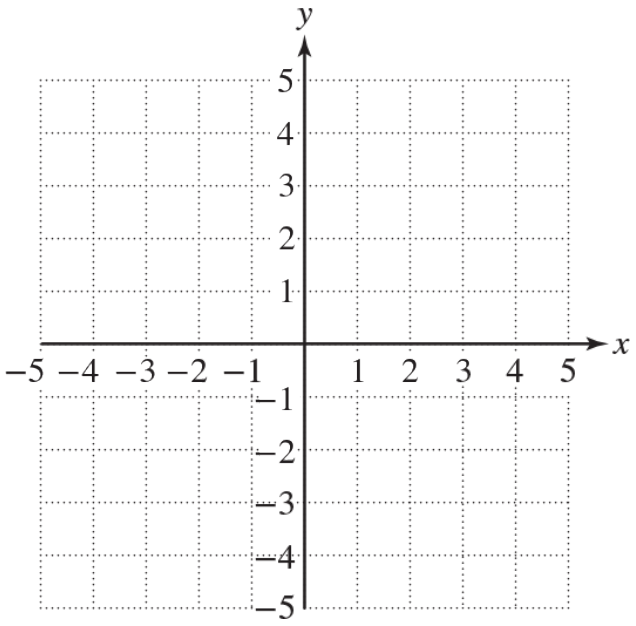
Note: A dashed line indicates that the line is *not* included in the solution set. A solid line implies that the line *is* included in the solution set.

Graph the solution set.

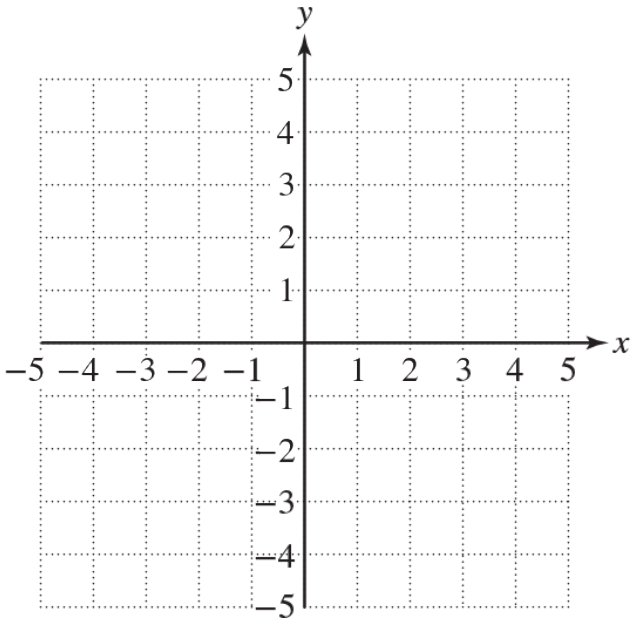
1. $-4y < 5x$



2. $5x - 2y < 10$



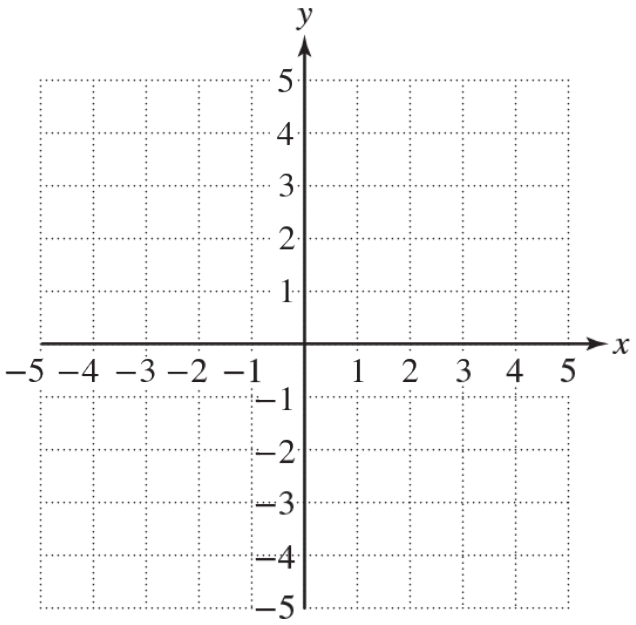
3. $2x \leq -6y + 12$



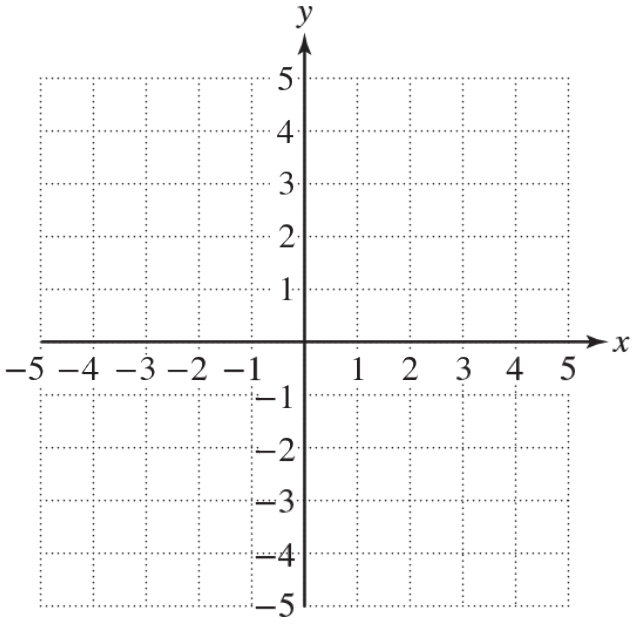
2 Systems of Linear Inequalities in Two Variables

Graph the solution set.

1. $\begin{cases} x + y < 4 \\ 3x + y < 9 \end{cases}$



2. $\begin{cases} x + 3y \geq 3 \\ x \leq -2 \end{cases}$



3. $\begin{cases} x \leq 1 \\ y > 2 \end{cases}$

